

A silicon image sensor that can compute



SEAS researchers developed the first in-sensor processor that could be integrated into commercial silicon imaging sensor chips. The array (illustrated here) simplifies image processing for autonomous vehicles and other applications (Credit: <https://techxplora.com>)

For autonomous vehicles to be as efficient and safe as a skilled driver, the system needs to develop its skill of analysing the environment and acting accordingly. The separate microprocessor used for image-processing in the autonomous vehicle can be replaced by an in-sensor silicon image processor to instantaneously compute the situation. To do this, researchers at the Harvard John A. Paulson School of Engineering and Applied Sciences have developed an electrostatically doped photodiode array. The array converts the incoming light directly into voltages. It connects multiple voltage-tunable photodiodes to perform operations in several imaging pipelines, thereby extracting the relevant visual information as soon as the image is captured. This imaging device can intelligently trim out unneeded data as well.

Smart hearing aids capable of lip reading

Hearing aids were capable of only helping hearing-impaired people to listen to what was going around, but the traditional hearing aids were of no use in crowded and noisy environments. To overcome this problem, some researchers implemented cameras for lip reading, but this could risk the privacy of an individual. Hence, researchers at the University of Glasgow have developed cutting-edge sensing technology to read lips without affecting individuals' privacy, as they receive data by collecting only radio frequency, which does not require video footage. This new system can read lips even when the speaker is wearing a face mask, which could revolutionise hearing aids.

Conceptual illustration of the proposed lip-reading framework. The framework employs Wi-Fi and radar technologies as enablers of RF sensing based lip-reading (Credit: <https://techxplora.com>)

